

Exploring an Innovative Deep Learning Solution for acupuncture point Localization on the Weak Feature Body Surface of the Human Back

Shilong Yang, Yalan Li, Lingfeng Huang, Jing Liu, Yongsheng Teng, Hao Zou, Yaoqin Xie

Abstract—In current clinical practice, the localization of human acupuncture points relies extensively on the subjective experience of physicians. Therefore, despite being a crucial basic content of traditional Chinese medicine (TCM), acupuncture point localization has not been well expanded and promoted through intelligent means. Our goal is to explore an efficient and reliable solution for acupuncture point localization and recognition that addresses the shortcomings of subjectivity and standardization in this task. We focus on the weak feature body surface of the human back and propose an innovative approach that utilizes a deep learning network with a self-attention module for global extraction of image features. This methodology differs from common Convolutional Neural Networks (CNNs) which often lead to classification ambiguous in weak feature image tasks due to excessive cropping and scaling operations during feature extraction. Moreover, our self-constructed dataset of human back acupuncture points provides data support for model training. The localization task for the back acupuncture points of the subjects in the dataset strictly follows the national standard definition and is labelled by professional doctors of TCM to ensure data robustness and quality. Our preliminary experiments validate that our proposed network learns higher-quality

global image features, achieving an average accuracy of less than 1cm in the localization and recognition task of 84 acupuncture points on the back of the human body. Access to our method is available at <https://github.com/Sohyu1/AL-WFBS>

Index Terms—acupuncture point localization, key points prediction, Artificial intelligence, Transformer, Traditional Chinese Medicine

I. INTRODUCTION

WITH the rapid advancement of information technology, artificial intelligence (AI) has deeply penetrated various aspects of human life. The field of medicine is no exception, as the demand for healthcare services continues to grow, fostering the robust development of information technology. In this overarching trend, the integration of artificial intelligence with medical technology has become a crucial initiative. AI-assisted diagnostic and therapeutic devices leverage massive datasets for targeted training and simulate the diagnostic thinking and reasoning processes of medical professionals. By automating the processing and inference of input data, these devices effectively save human and material resources, providing robust support for clinical decision-making.

Despite the flourishing development of AI technology in the medical field, the unique nature of traditional Chinese medicine (TCM), distinct from the quantitative standardization of Western medicine, has resulted in a relatively lagging adoption of AI in TCM. While in Western medicine, doctors increasingly utilize AI-assisted devices for treatment and diagnosis, such adoption is less prevalent in TCM, where practitioners still heavily rely on their rich experience for differential diagnosis and treatment. In the realm of differential diagnosis and treatment in TCM, the lack of unified standards makes it challenging to derive appropriate data-driven methods from long-term clinical experiences.

The meridians and acupuncture points are indispensable components of traditional Chinese medicine, widely employed in various therapeutic interventions over the past three thousand years. Acupuncture, as a significant treatment modality, is not only prevalent in traditional Chinese medicine but has also been utilized in Western countries to treat various chronic conditions such as vomiting, hot flashes, migraines, and knee pain [1]. In situations where conventional treatments prove ineffective, many patients turn to acupuncture for potential relief [2], [3]. Modern scholars generally explain the clinical

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applicability of acupuncture through neuroscience, considering it an alternative therapy for inflammation and pain [4] [5] [6]. Some studies suggest that acupuncture can independently activate the body's self-healing mechanisms, restore homeostasis, effectively treat immune diseases, and be applied to mental disorders such as depression and Parkinson's disease [7] [8] [9] [10]. Moxibustion, a form of acupuncture, is a traditional Chinese medical technique that has experienced robust development due to its simplicity, cost-effectiveness, and notable efficacy. With the rapid advancement of computer technology and the continuous emergence of various machine learning and deep learning algorithms, the advent of machine vision systems has transformed the control processes of robotic arms. Through machine vision systems, the precise localization of target objects is achieved, significantly enhancing safety and accuracy. Robotic arms have entered an era of automated, real-time, and precise control, during this time, various medical robots have proliferated. In the current era of flourishing intelligent medical devices, an automatic moxibustion device that achieves automatic localization of acupuncture points holds significant implications for traditional Chinese medicine treatment.

Due to the high accuracy requirements of moxibustion for acupuncture point localization, it is essential to identify the precise locations of acupuncture points and calculate the appropriate distance from the moxibustion stick to the skin surface to ensure the effectiveness of treatment. During actual TCM treatments, back moxibustion often cannot be independently performed, and doctors typically rely on their own experience to locate acupuncture points. In moxibustion, the burning position of the moxibustion stick needs to maintain a stable distance from the acupuncture point to avoid the risk of skin burns. Therefore, to ensure the safety of the moxibustion device, achieving accurate three-dimensional spatial localization of acupuncture points is crucial. We propose a real-time three-dimensional acupuncture point prediction method that utilizes RGB-D cameras to obtain real-time images and depth images of the patient's back, allowing for precise localization of acupuncture points.

Our work represents the first application of key point recognition for predicting torso acupuncture points, utilizing transformers to forecast corresponding points. Our primary contributions include:

- We have developed a network for predicting acupuncture point locations, enabling precise measurement and localization. This data-driven method, powered by artificial intelligence, significantly improves the accuracy and objectivity of acupuncture point localization.
- We incorporated a key point recognition module with the RT-DETR (Real-Time Object Detection with Transformers) target recognition network. This network not only retains the high accuracy of target recognition from RT-DETR but also predicts key points based on the identified targets, providing enhanced constraints for precise key point localization.
- Furthermore, following national standards, we have established the inaugural RGB-D dataset for back acupuncture points. This dataset comprises back images of multiple

subjects captured in a controlled environment, along with annotations for 86 corresponding acupuncture point locations on the back.

II. RELATED WORK

A. The Current State of Research in Image Recognition

Human perception of the world is predominantly facilitated by the five senses, with visual information accounting for 80% of the total input. Analogous to the principles of human vision, machine vision involves the acquisition of images, followed by a series of processing and analysis steps, such as feature extraction, recognition, and classification. These processes are integral for practical applications in detection and control [11]. As science and technology continually advance, image target recognition methods in machine vision are increasingly finding broad applications in academic research fields, including but not limited to unmanned driving, medical diagnosis, and facial recognition [12].

In the early stages of development, prevalent target recognition algorithms were typically based on classifier methods, also known as statistical-based target recognition. Notable examples of classifier-based target recognition algorithms include the Adaboost algorithm, the Support Vector Machine (SVM) learning algorithm, and the neural network algorithm. The Adaboost iterative algorithm [13], introduced by Freund et al. in 1995, involves training diverse weak classifiers on the same training set, combining them with assigned weights to construct a final robust classifier. This algorithm's appeal lies in its simplicity and efficiency in weak classifier construction, ultimately achieving high classification accuracy. In the same year, Cortes and Vapnik proposed the SVM algorithm [14], essentially a linear classifier maximizing the margin in feature space. This algorithm is particularly suitable for scenarios involving small samples, non-linearity, and high dimensions, providing a classifier with high generalization ability, low empirical risk, and low confidence risk.

With the evolution of deep learning technology, numerous target recognition algorithms based on neural networks have emerged. In 2014, Ross Girshick integrated region proposals with CNNs to propose the RCNN (Regions with CNN features) algorithm, transforming the object detection problem into a classification challenge. This approach aimed to fully leverage the advantages of convolutional neural networks (CNNs) in classification tasks. The algorithm utilized region proposal and region-based convolutional neural networks to infer target positions; however, the computational cost associated with region proposals was significant and time-consuming. Building on the RCNN in 2015, Ross Girshick proposed Fast-RCNN [15], which mapped proposed regions only to the last layer of the CNN's feature map, thus enhancing recognition speed and accuracy. In the same year, researchers, including He Kaiming and Ross Girshick, continued to refine Fast-RCNN, leading to the introduction of Faster R-CNN, further improving operational speed.

In 2016 researchers from UNC Chapel Hill proposed the SSD (Single Shot Multibox Detector) model [16] which streamlined the detection process into a single deep neural

network. This approach facilitated ease of training and optimization, ensuring a balance between speed and accuracy. Currently, deep learning stands as a focal point of interest for many researchers. It is anticipated that with continued research, the computational speed limitations of deep learning will be surmounted, propelling it to new heights in the miniaturization and practical application of target recognition in engineering.

In their study, Fang *et al.* [17] achieved three-dimensional object recognition by extracting grid-based statistical model features from two-dimensional images at various viewpoints of the target. This approach not only enabled the identification of three-dimensional objects but also, in conjunction with a model library, facilitated the estimation of the three-dimensional object poses. Addressing target recognition in a battlefield environment, Li *et al.* [18] proposed the projection of the target's three-dimensional model onto a two-dimensional plane. They utilized the cosine Fourier moment and Rayleigh distribution in a Constant False Alarm Rate (CFAR) detection method to extract moment and peak features, respectively. The researchers employed a cascaded ensemble classifier for effective target recognition.

B. The Current State of Research in Visual Localization Technology

Visual localization refers to the process of determining the position information of a target from images. This theory integrates data information collected from cameras and other sensors. Visual localization methods are primarily categorized into monocular, binocular, and multi-camera visual localization methods. Monocular visual localization eliminates the need to address issues such as optimal distance between the two cameras in binocular vision and feature point matching. It also avoids dealing with significant distortion between multiple cameras.

On the other hand, binocular (stereo) and multi-directional (omnidirectional) visual localization methods are challenging to implement, often exhibiting lower accuracy and real-time performance, making them unsuitable for meeting the precision requirements of industrial robot processing. In general, when using monocular visual localization methods, key factors affecting the accuracy include the camera's inherent performance, algorithmic computational complexity, and the precision of feature point extraction and matching. Addressing these challenges is crucial for advancing industrial robot visual localization accuracy.

Currently, visual localization is often combined with key point recognition. Key point detection plays a crucial role in computer vision, encompassing various subfields such as facial key point detection, hand key point detection, and human body key point detection. Facial key point detection focuses on automatically locating a set of predefined reference points on the face, including the corners of the eyes, the centre of the eyebrows, the corners of the mouth, and the tip of the nose. This technology has found widespread applications in areas like facial recognition and facial editing, making significant strides in recent years.

Hand key point detection and human body key point detection aim to identify the positions of joints in images, facilitating various subsequent tasks such as human-computer interaction, pose estimation, and behaviour recognition. Convolutional Neural Networks (CNNs) have demonstrated exceptional learning and generalization capabilities, making significant strides in various fields, including key point recognition.

Researchers [19]–[24], among others, have successfully utilized CNNs to achieve outstanding performance in key point detection tasks.

Currently, key point detection methods include various approaches such as direct coordinate regression heatmap regression and heatmap with offsets. Toshev and Szegedy [19] employed a cascaded detector to refine poses in a top-down manner from cropped image patches and directly returned key points. Li *et al.* [25] employed Residual Log-likelihood Estimation and the flow method to successfully estimate the probability density distribution of joint outputs in the model. Once a satisfactory prior distribution function is obtained, it becomes possible to dynamically optimize the loss function, thereby promoting the regression training of the model. This marks the first instance where regression-based methods outperform heatmap-based methods in terms of performance, and they are also more computationally and storage-efficient.

Jain *et al.* [20] proposed an end-to-end model for two-stage human pose estimation based on convolutional networks enabling CNNs to work effectively in the pose estimation domain. They initially trained multiple binary classification convolutional networks for different body parts, generating a separate heatmap for each key point, indicating confidence in the presence of a body part at that location. They then inferred the human pose based on the detected key points. To further improve results and ensure consistency between key points, they introduced a "DPM-like" graphical model post-processing step. This step involved filtering potential key points and reinforcing relationships between key points, enhancing the accuracy and reliability of the key point estimation process. HRNet [26] is a parallel structure to simply and robustly connect sub-networks of different resolutions. To address the issue of small receptive fields in high-resolution representations, HRNet progressively increases the resolution from high to low by adding sub-networks, thereby acquiring more global information. Through a feature fusion module, high-resolution representation information is exchanged with low-resolution representation information, enhancing the learning of high-resolution representations with low-resolution features. Simultaneously, high-resolution representations focus on local information features to strengthen the learning of global low-resolution representations. The VitPose [27] was proposed based on the Vision Transformer, renowned for its straightforward, adaptable, and versatile architecture. It resulted in an outstanding performance on the COCO dataset [28], setting a new benchmark.

In 2017, Google introduced G-RMI [21], a novel method aimed at addressing the resolution reduction issue caused by downsampling in traditional heatmap methods. G-RMI combines heatmap and offset techniques to enhance the accuracy of keypoint localization. By incorporating two-dimensional

offsets generated by the network, G-RMI achieves higher precision in determining key point positions.

Differing from the aforementioned three methods, SimCC [29] introduces a novel approach that treats keypoint localization as a classification task. It represents keypoint prediction as the classification of sub-pixel boxes along both horizontal and vertical coordinates. This approach brings several advantages. Firstly, SimCC eliminates the dependence on high-resolution heatmaps and costly upsampling. It flattens the final feature map for classification, avoiding global pooling [19] and preventing the loss of spatial information. Thirdly, the classification of coordinates at sub-pixel scales effectively mitigates quantization errors without the need for additional post-processing [30]. In comparison to heatmap-based algorithms, SimCC-based algorithms achieve higher accuracy with a lower computational workload.

C. The Current Research Status of Acupuncture Point Localization

The earliest exploration of automatic acupuncture point localization can be traced back to 2011 when researchers, including Du et al. [31], attempted to integrate traditional Chinese massage robots with acupuncture point localization. Subsequently, other researchers used traditional methods to achieve acupuncture point localization. Li et al. [32] designed a head and neck acupuncture point measurement and localization system that utilized optical instruments and image processing methods to project acupuncture points. Researchers at Shandong Jianzhu University, led by Zhang [33], developed a traditional Chinese massage robot capable of tracking and locating acupuncture points based on templates, earning recognition at the provincial level.

In recent years, with the rapid development of deep learning technology, researchers have begun applying deep learning to the field of acupuncture point localization. Gong et al. [34] proposed a mobile acupuncture point localization software that scales annotated acupuncture point images to the size of a human body using projection devices, projecting them onto the body surface to achieve acupuncture point localization. Wang from Beijing University of Posts and Telecommunications [35] constructed a visual acupuncture point localization system using multiple depth cameras. The system first recognized the human body posture using a global camera, then localized acupuncture points on specific limbs using a local camera. Finally, it employed optical flow to track feature points, achieving accurate and effective relocalization of acupuncture points in response to limb movements, effectively addressing the acupuncture point localization challenges in laser acupuncture robots. Zheng [36] utilized deep learning to develop a mobile AR system capable of automatically identifying acupuncture points on the face in real-time. In Table I, we compare various studies on acupuncture point localization.

As acupuncture point localization is currently in the research stage and has limited clinical applications, there is a scarcity of data for research, posing certain difficulties for researchers.

III. METHOD

In this section, the proposed AL-WFBS(acupuncture point Localization on the Weak Feature Body Surface) network structure is described in detail and a framework for acupuncture point prediction is established. In section 3.1, we introduced the experimental dataset and preprocessing procedure. In section 3.2, we formulated the acupuncture point registration problem mathematically. In section 3.3, we provided a detailed introduction to the AL-WFBS framework. Section 3.4 shows the implementation details.

A. Data set and processing

The dataset of back acupuncture points was collaboratively collected through a partnership between our team and Traditional Chinese Medicine (TCM) practitioners from Shenzhen General Hospital. The physicians were responsible for manual annotation, while we managed the digital acquisition process. We recruited nearly 200 participants, all of whom signed informed consent forms. These participants were healthy individuals without apparent spinal deformities and with normal body shapes. Here, the definition of body shape is determined by the Body Mass Index (BMI) calculation. Before the experiment, we obtained the current height and weight of all subjects through a questionnaire and calculated the BMI results accordingly, using the following formula.

$$BMI = \frac{weight(kg)}{(height(m))^2} \quad (1)$$

Table II shows that our data covers all BMI categories, with the majority being of normal body type.

TABLE II
PROPORTION OF EACH BMI CATEGORY IN THE DATASET.

	underweight	normal	overweight	obesity
BMI	0.05	0.77	0.16	0.02

Three experienced TCM doctors identified and recognized the back acupuncture points based on the Chinese national standard "Names and Locations of acupuncture points." The identified points were marked, and the participants, lying prone on a treatment bed, were then photographed using a depth camera to capture corresponding RGB and depth images of their backs. Subsequently, the annotations were carried out using labelme software.

The data collection process consists of four steps:

- Volunteers will fill out an informed consent form and provide necessary personal information, including name, gender, age, height, and weight.
- Subjects will sit upright with arms naturally hanging down. A professional TCM practitioner will identify and recognize acupuncture points on the subject's back. Female subjects will wear undergarments, and all staff will be female. Mark stickers will be used to label identified acupuncture points.
- Subjects will lie supine on a treatment bed. For female subjects, all staff will be female, and they will wear undergarments. A depth camera will capture images,

TABLE I
COMPARATIVE ANALYSIS TABLE OF RESEARCH ON ACUPUNCTURE LOCALIZATION DIRECTION.

	Kong's	Wang's	Chen's	our
Positioning accuracy	low	high	medium	high
convenience	✓	×	✓	✓
cost	low	high	low	low
scope of application	widely	limit	limit	widely
technical requirement	low	high	low	low

focusing only on the back, to obtain RGB images with marked points.

- Mark stickers will be removed, and subjects will maintain the same position. The back will be photographed again to obtain images without marks.

The experiment will be conducted entirely in a shielded room, with closed data collection throughout to ensure privacy and security. During our data creation process, staff will use Labelme to individually annotate each volunteer's data and ultimately save it as a JSON file. Throughout this process, strict guidelines are enforced to prevent information leakage and address privacy and ethical concerns.

In total, there were 84 acupuncture points on the back, spanning four meridians. Fig.1 illustrates the locations and names of these 84 points as outlined in the "Names and Locations of acupuncture points." The entire process involved meticulous collaboration between TCM professionals and digital acquisition experts to ensure accurate and comprehensive dataset creation.

Due to challenges faced during the collection process for female participants, the number of female samples in our dataset is significantly lower than that of male samples, resulting in a dataset gender ratio of 3:1.

During the actual training process, we applied data augmentation techniques to enhance the dataset. Subsequently, we randomly shuffled and partitioned the augmented dataset into training, validation, and test sets, following a split ratio of 81:9:10. This approach ensured a diverse and well-distributed dataset for effective model training, validation, and testing.

B. Data Security and Experimental Ethics

We provided informed consent forms to all volunteers, allowing them to clearly understand the purpose of data use, methods of collection, and potential risks and benefits, ensuring voluntary participation. The collected data is coded to replace real identity information, and the collection sites do not involve the volunteers' faces or private areas.

All volunteers will complete a simple questionnaire that gathers only the minimal data necessary to achieve the research objectives, avoiding unnecessary information collection. Regarding data security, we implement strict security measures, including access control and regular security audits, to limit data sharing and access. Only authorized personnel can access the data when necessary, protecting it from unauthorized access and breaches. Additionally, our research protocol has been submitted to and approved by the ethics committees of relevant institutions, ensuring the study design meets ethical standards.

C. Algorithm description

The problem of acupuncture point localization shares a certain resemblance with key point detection. A viable approach involves training a keypoint detection network, utilizing the Euclidean distance between ground truth and predicted values as the loss function. Essentially, this task boils down to addressing a key point detection problem.

In practical terms, given a completely prone back image, we employ a keypoint detection network to predict the locations of acupuncture points. This process essentially transforms the task of pinpointing acupuncture locations into a keypoint detection challenge, where the network's objective is to accurately identify and locate specific key points on the back. This approach aligns with the broader paradigm of addressing key point detection challenges in medical image analysis.

For a given input image p , coordinates (x, y) are predicted, where $x, y \in [0, 1]$. The pixel distance loss between these predicted coordinates and the corresponding true coordinates (x', y') , where $x', y' \in [0, 1]$ also ranges from 0 to 1, is calculated. The objective is to find an optimal set of parameters such that the loss between the transformed coordinates (x, y) after applying the spatial transformation function $T_\phi(P)$ and the true coordinates (x', y') is minimized.

$$\arg \min_{\phi} \mathbb{E}_{((x', y'), P)} L_P \quad (2)$$

$$L_p = \frac{\|(x', y') - T_\phi(P)\|^2}{N_{p'}} \quad (3)$$

D. Network Structure

In Fig.2, we illustrate the network architecture of AL-WFBS. Our model is built upon the foundation of the DETR network architecture, integrating heatmap regression with the Transformer model. Leveraging the advantages inherited from attention modules, our approach computes the mean square loss between predicted points and ground truth points, enabling accurate prediction of acupuncture point positions. This integration highlights the synergy between key point regression and the Transformer model, showcasing the effectiveness of our network in precise acupuncture point localization.

The proposed AL-WFBS model encompasses a CNN backbone, hybrid encoder, transformer decoder, and auxiliary prediction heads. Specifically, our approach involves feature extraction from input images using a CNN backbone, with the extracted output features from three stages (S3, S4, S5) serving as input to the encoder. The primary function of the hybrid encoder is to convert multiscale features into a series of image features by facilitating interscale interaction

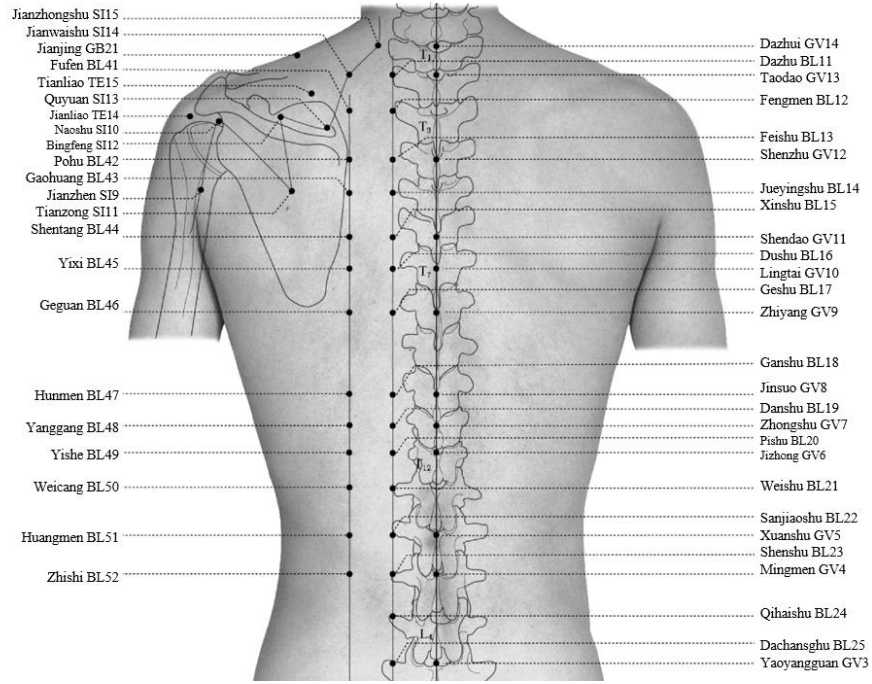


Fig. 1. Localization guideline of acupuncture points.

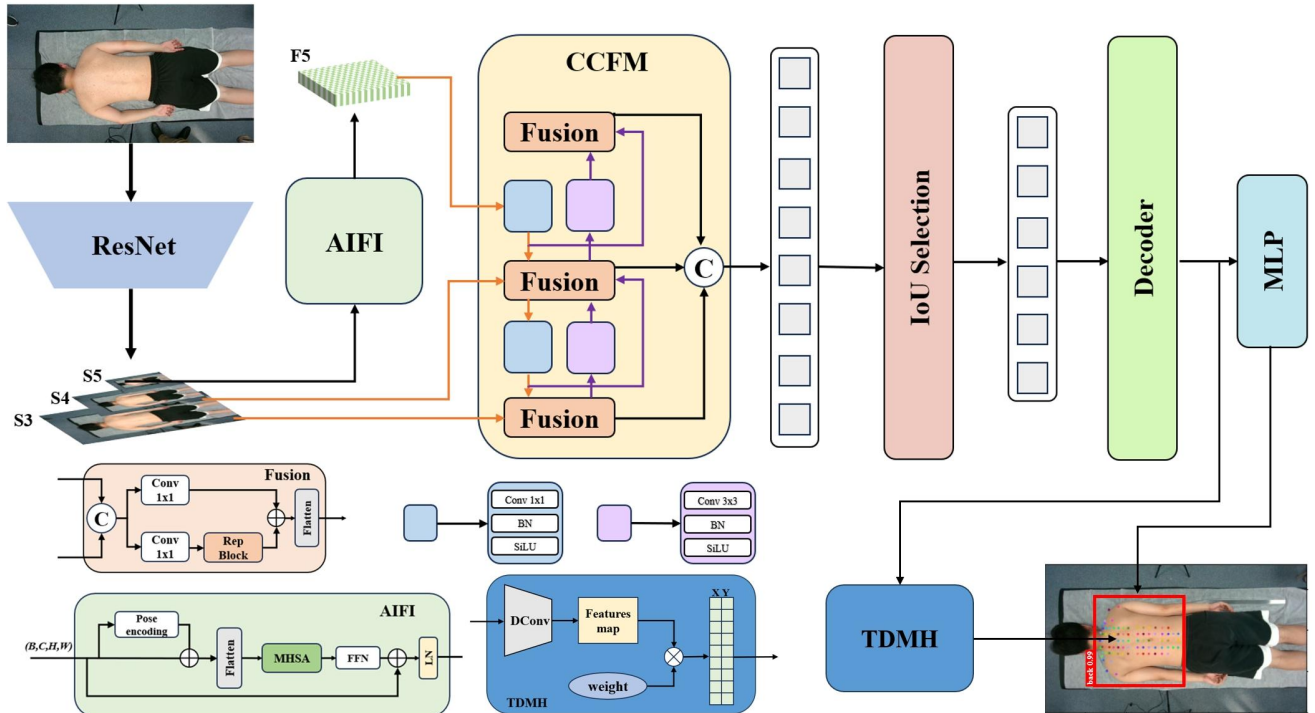


Fig. 2. Schematic of the AL-WFBS.

and cross-scale fusion. Subsequently, we introduce an IOU-aware query selection mechanism, selecting a fixed number of image features from the encoder's output sequence to serve as initial object queries for the decoder. Lastly, the decoder, in collaboration with auxiliary predicts heads, iteratively refines these object queries. Subsequently, the output results are passed through both an MLP layer and a top-down multi-head heatmap module, achieving regression for the categories and positions in object detection, and predicting key points. This approach represents an innovative solution for transforming multiscale features into actionable object predictions.

E. Implementation Details

We employ ResNet-50 [37] as the backbone network for local feature extraction, initializing it with the weights pre-trained on ImageNet [38]. Subsequently, we extract outputs at three scales with strides of 8, 16, and 32, denoted as S3, S4, and S5, respectively. We transform the 2D S5 into a vector and feed it individually into the encoder module, then reshape the output back to 2D, denoted as F5. The obtained F5, along with S3 and S4, is input into PaFPN, where several layers of 1x1 convolutions map the channel dimensions of all features to a common number. Top-down and bottom-up feature fusion is then performed, achieving multiscale feature map integration.

In the subsequent IOU-aware query selection, we choose the top 300 encoder features to initialize the decoder's object queries. The training strategy and hyperparameters of the decoder closely follow DINO [39]. We opt for the Adam optimizer [40], initially freezing the pre-trained backbone network with a learning rate of 10^{-4} , batch size of 1, and a total iteration count of 300,000. Subsequently, we unfreeze the backbone network, fine-tuning the feature extraction network with a learning rate of 10^{-5} and batch size of 1 to enhance its relevance to our dataset.

The experiments were conducted on a single NVIDIA GPU (GeForce RTX 3090 12GB), utilizing the PyTorch framework [41]. The training process took approximately 5 hours.

In view of the high-frequency jitter of key points that often occur in real-time detection, this situation greatly affects the visual perception of real-time display. To solve this problem, we use 1€ filter [42] to filter the output of the attitude estimation model. This measure has significantly improved the visual perception of the filtered results and has significantly improved the visualization effect of real-time detection.

F. Evaluation Indicator

We propose two evaluation metrics for the accuracy of acupuncture point prediction, called Acupuncture-points Prediction Error (APE) and Relative Acupuncture-points Prediction Error (rAPE). We define them as follows:

$$APE(T_\phi, (x_j', y_j'), P) = \frac{1}{n} * \sum_{(x_j, y_j) \in n} ((x_j', y_j') - T_\phi(P))^2 \quad (4)$$

$$rAPE(T_\phi, (x_j', y_j'), P) = \frac{1}{n} * \sum_{(x_j, y_j) \in n} \frac{((x_j', y_j') - T_\phi(P))^2}{N} \quad (5)$$

n represents the number of acupuncture points in the test, given a transformation function T_ϕ , this problem can be described as calculating the mean square distance from (x_j, y_j) to (x_j', y_j') after T_ϕ by traversing the point on each image.

"rAPE" is a normalized version of the "APE" metric that takes into account the scale factor of different images. Here, "N" represents the scale factor of the test image, which is the Euclidean distance between the middle of the Dazhui acupuncture point and the Yaoyangguan acupuncture point on the upper Du vein. "Average Acupuncture-points Prediction Error (AAPE)" and "Average Relative Acupuncture-points Prediction Error (ArAPE)" represent the average "APE" and "rAPE" values across the test samples respectively.

The Percentage of Correct Key points (PCK) is a metric that reports the percentage of correctly detected key points that fall within a normalized distance from the ground truth.

$$PCK_i^k = \frac{\sum_n \delta\left(\frac{d_i}{N_j} \leq H_k\right)}{\sum_n 1} \quad (6)$$

$$PCK_{mean}^k = \frac{\sum_n \sum_i \delta\left(\frac{d_i}{N_j} \leq H_k\right)}{\sum_n \sum_i 1} \quad (7)$$

Here, i represents the query point with ID i , k represents the k th threshold, n represents the number of queries pairs in the test, d_i represents the Euclidean distance between the predicted value of the i th point and the ground truth, N_j represents the scale factor of the j th image pair, H_k represents the given threshold, where $H_k \in [0.05 : 0.10 : 0.15]$, PCK_i^k represents the PCK metric for query point i under threshold H_k , and PCK_{mean}^k represents the PCK metric under threshold H_k averaged across all query points.

IV. RESULT

A. Prediction Results

In our network architecture, the training output encompasses the predicted coordinates of all acupuncture points in each test image. These predicted coordinates, after undergoing a meticulously designed coordinate transformation strategy, provide us with accurate and detailed results. It is noteworthy that, compared to other methods, our training process requires relatively fewer epochs while achieving excellent performance. This reflects the efficiency and superior capabilities of our model. To visually showcase the performance of AL-WFBS, we present the predicted results of the test set in Fig.3. In Fig.3, green represents the actual distribution of acupuncture points (ground truth), while red signifies our predicted results. This illustration clearly demonstrates the performance of AL-WFBS on test data, highlighting the model's accuracy and reliability.

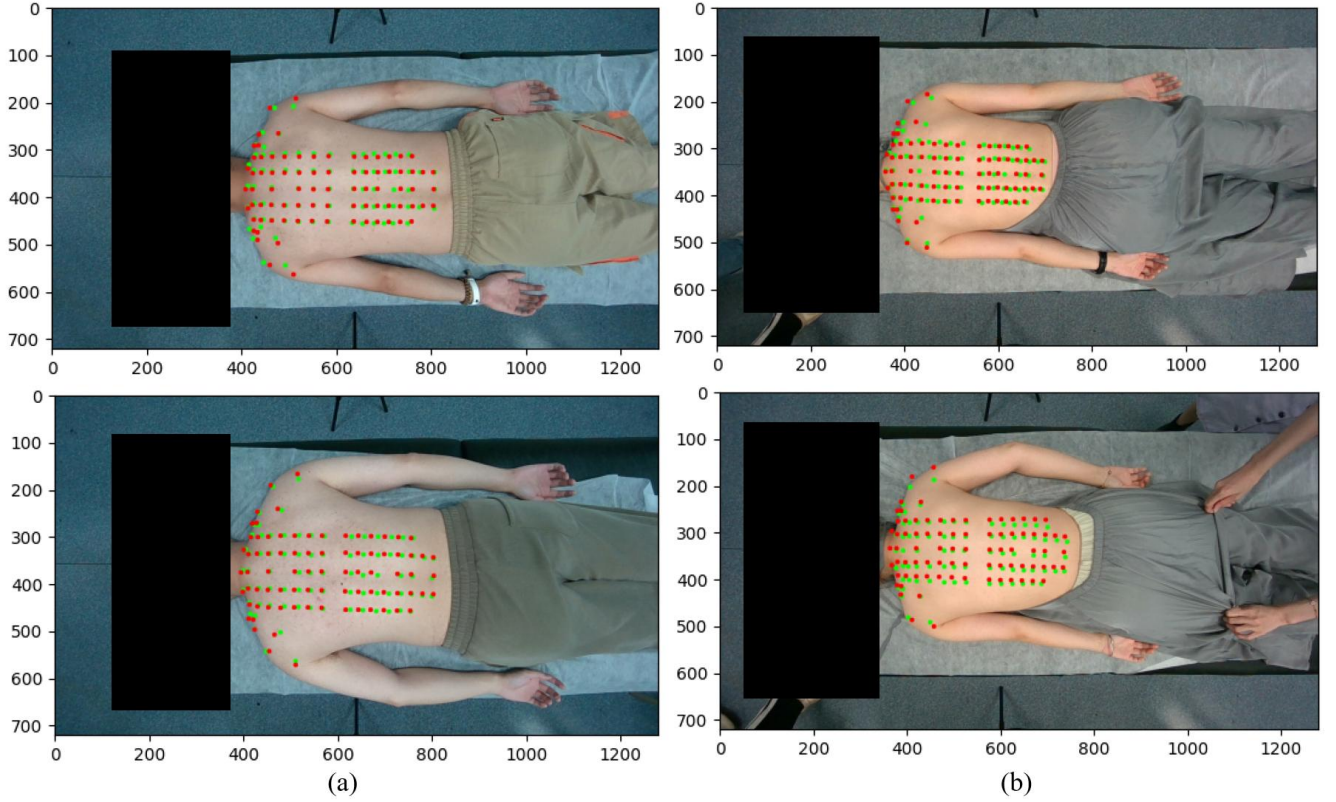


Fig. 3. AL-WFBS's prediction results on the test set are ground-truth in green and prediction results in red.

From Fig.3, it can be observed that AL-WFBS's predictions are quite accurate on the Governing Vessel and Bladder Meridian, but there is a slight deviation in the predictions for the shoulders. Table III presents the performance results of our network on an equal number of male and female test sets. The PCK and AAPE in the male test set are superior to those in the female test set. We speculate that this is due to a higher proportion of male data in the training dataset, leading to more accurate predictions for males compared to females.

TABLE III
PERFORMANCE EVALUATION OF AL-WFBS ON MALE AND FEMALE TEST SETS

gender	AAPE	PCK-0.05	PCK-0.10
Male	9.11	0.89	1.00
Female	11.35	0.89	0.98

Given that the cost of simultaneous inference may be related to the number of acupuncture points predicted each time, and predicting multiple acupuncture points at the same time incurs higher costs than predicting only one acupuncture point, reducing the number of acupuncture points predicted each time is a cost-reduction strategy. The personalized positioning of traditional Chinese medicine acupuncture points strictly follows the principles of bone measurement. The location of acupuncture points along the Bladder Meridian on the back strictly depends on the position of acupuncture points along the Governor vessel meridian. Some of the existing acupuncture point prediction methods only predict the position of the

benchmark acupuncture points, and the rest of the acupuncture points are deduced according to the formula. Imitating [43], we can set the acupuncture points along the Governing Vessel as the baseline for positioning, while other acupuncture points depend on individual measurements and the body's angular displacement to calculate, shoulder acupuncture points are still directly predicted. In our experiments, we selected the acupuncture points Dazhui, Yaoyangguan, Dazhu, and Taodao as the baseline acupuncture points. Among them, Dazhui and Yaoyangguan are used to calculate the body's inclination angle θ , while Dazhu and Taodao are used to calculate the pixel distance l corresponding to one cun of the body. The formula for calculating the body's inclination angle θ and one cun l are as follows.

$$\sin \theta = \frac{y_b - y_a}{\sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}} \quad (8)$$

$$\cos \theta = \frac{x_b - x_a}{\sqrt{(x_b - x_a)^2 + (y_b - y_a)^2}} \quad (9)$$

$$l = \frac{2}{3} \times \sqrt{(x_c - x_d)^2 + (y_c - y_d)^2} \quad (10)$$

The coordinates of Dazhui Point, Yaoyangguan Point, Dazhu Point and Taodao Point are (x_a, y_a) , (x_b, y_b) , (x_c, y_c) and (x_d, y_d) . Some acupuncture point correction calculations are shown in Table IV.

Fig.4 and Table V compare the results of these two workflows. Since the shoulder acupuncture points were predicted,

TABLE IV
 PART OF THE ACUPUNCTURE POINT CORRECTION CALCULATION TABLE

acupuncture point name	x-coordinate	y-coordinate
Dazhui	X	Y
Shenzhu	$X + 5l \cos \theta$	$Y + 5l \sin \theta$
Shendao	$X + 7.5l \cos \theta$	$Y + 7.5l \sin \theta$
Lingtai	$X + 8.75l \cos \theta$	$Y + 8.75l \sin \theta$
Zhiyang	$X + 10l \cos \theta$	$Y + 10l \sin \theta$
Jingsuo	$X + 12.67l \cos \theta$	$Y + 12.67l \sin \theta$
Zhongshu	$X + 14l \cos \theta$	$Y + 14l \sin \theta$
Jizhong	$X + 15l \cos \theta$	$Y + 15l \sin \theta$
Xuanshu	$X + 17l \cos \theta$	$Y + 17l \sin \theta$
Mingmen	$X + 18l \cos \theta$	$Y + 18l \sin \theta$
Fengmen	$X + 3.25l \cos \theta \pm 1.5l \sin \theta$	$Y + 3.25l \sin \theta \pm 1.5l \cos \theta$
Feishu	$X + 5l \cos \theta \pm 1.5l \sin \theta$	$Y + 5l \sin \theta \pm 1.5l \cos \theta$
Jueyin	$X + 6.25l \cos \theta \pm 1.5l \sin \theta$	$Y + 6.25l \sin \theta \pm 1.5l \cos \theta$
Gaohuang	$X + 6.25l \cos \theta \pm 3.0l \sin \theta$	$Y + 6.25l \sin \theta \pm 3.0l \cos \theta$
Xinshu	$X + 7.5l \cos \theta \pm 1.5l \sin \theta$	$Y + 7.5l \sin \theta \pm 1.5l \cos \theta$
Shentang	$X + 7.5l \cos \theta \pm 3.0l \sin \theta$	$Y + 7.5l \sin \theta \pm 3.0l \cos \theta$
Dushu	$X + 8.75l \cos \theta \pm 1.5l \sin \theta$	$Y + 8.75l \sin \theta \pm 1.5l \cos \theta$
Geshu	$X + 10l \cos \theta \pm 1.5l \sin \theta$	$Y + 10l \sin \theta \pm 1.5l \cos \theta$
Geguan	$X + 10l \cos \theta \pm 3.0l \sin \theta$	$Y + 10l \sin \theta \pm 3.0l \cos \theta$
Ganshu	$X + 12.67l \cos \theta \pm 1.5l \sin \theta$	$Y + 12.67l \sin \theta \pm 1.5l \cos \theta$
Danshu	$X + 14l \cos \theta \pm 1.5l \sin \theta$	$Y + 14l \sin \theta \pm 1.5l \cos \theta$
Pishu	$X + 15l \cos \theta \pm 1.5l \sin \theta$	$Y + 15l \sin \theta \pm 1.5l \cos \theta$
Weishu	$X + 16l \cos \theta \pm 1.5l \sin \theta$	$Y + 16l \sin \theta \pm 1.5l \cos \theta$
Weicang	$X + 16l \cos \theta \pm 3.0l \sin \theta$	$Y + 16l \sin \theta \pm 3.0l \cos \theta$
Sanjiao	$X + 17l \cos \theta \pm 1.5l \sin \theta$	$Y + 17l \sin \theta \pm 1.5l \cos \theta$
Shenshu	$X + 18l \cos \theta \pm 1.5l \sin \theta$	$Y + 18l \sin \theta \pm 1.5l \cos \theta$

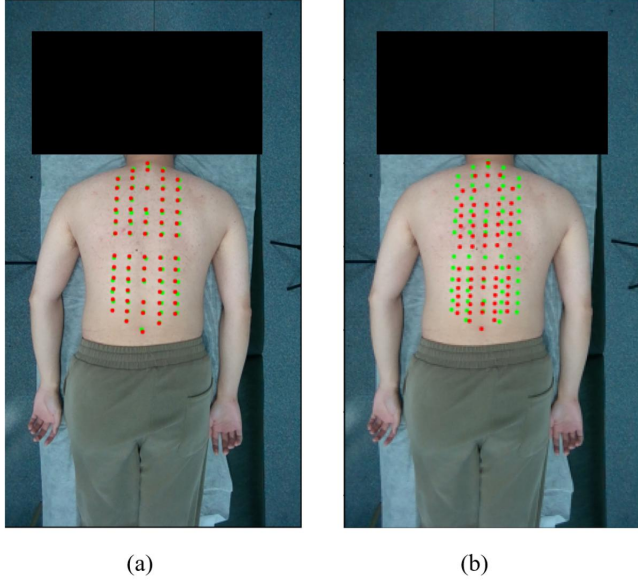


Fig. 4. A Comparative Analysis of Acupuncture Point Predictions under Two Workflow Scenarios: This figure compares the results of two workflow scenarios—one directly predicting all acupuncture points(a) and the other inferring predictions for non-anchor acupuncture points based on anchor acupuncture points(b). Ground truth values (in green) and predicted results (in red) are highlighted.

 TABLE V
 THE RESULT OF TWO WORKFLOWS

Workflow type	84 points	formula calculation
Predicted speed(ms)	72.70	62.45
Accuracy(AAPE)	9.29	26.15

we only compared the two workflows for the acupuncture points of the governor vessel meridian and the bladder meridian. Fig.4 only shows the acupuncture point results of the governor's vein and the bladder meridian. From a temporal perspective, the time cost of relying on baseline acupuncture points for prediction is only slightly reduced compared to the proposed AL-WFBS. However, from the standpoint of prediction accuracy, the accuracy of the baseline acupuncture point-based localization method is noticeably lower than our method. Considering the decrease in accuracy, relative to the reduction in time cost, our proposed method is much more practical than the baseline acupuncture point-based localization.

B. Comparison to Existing Method

We assess our method and state-of-the-art key point detection approaches under identical configurations, employing the same hardware settings and data preprocessing procedures. The evaluation employs test datasets, where we compute the Acupuncture-points Prediction Error (APE) for all valid pixels and the Percentage of Correct Key points (PCK) at specified reprojection error thresholds (0.05, 0.10). Given that acupuncture points in traditional Chinese medicine theory are conceptualized as circular regions with a diameter of 1 cm rather than pinpoint-sized dots, any point falling within this circular area is considered a correct coordinate. To address scale discrepancies caused by varying distances from the human body to the camera, we adhere to the principle of "same inch in Chinese medicine theory." The actual distance from Dazhui to Yaoyangguan is calculated for each image, setting this as the image's scale factor, and normalising the evaluation metrics accordingly. The summarized results are presented in Table VI. It can be observed that certain methods

exhibit relatively high APE values while maintaining high PCK values. This discrepancy arises from inaccuracies in predicting key points, causing these points to be far from the true coordinate points and consequently inflating the average pixel error.

We validated our model's effectiveness by conducting comparisons with other models and assessing their performance using five standard metrics: AP, AR, GFLOPs, Parameters, and FPS. GFLOPs and Parameters evaluate computational complexity and storage needs, while FPS indicates inference speed. The summarized results are presented in Table VII.

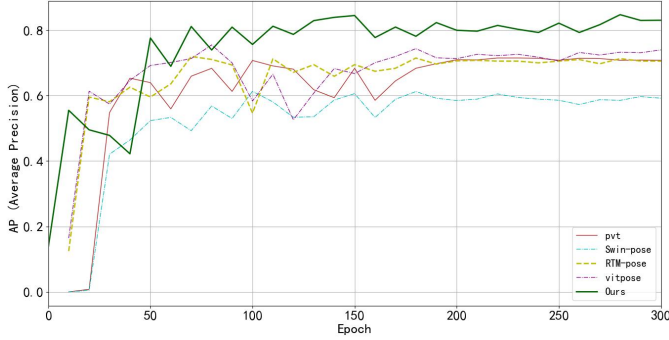


Fig. 5. The AP curves of the validation set.

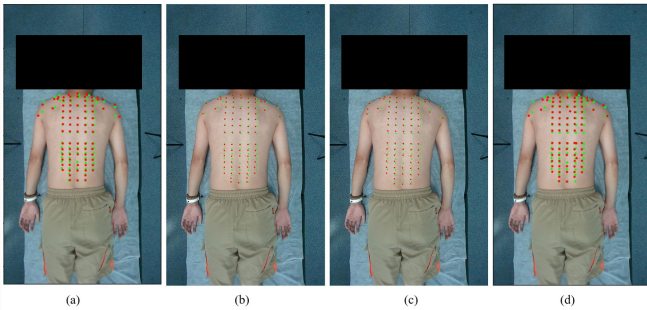


Fig. 6. Our method, RTM-pose, VitPose, and YoloV8 use the same configuration to train 300 epochs to predict the same image, ground truth values (in green) and predicted results (in red). (a)our method ,(b)RTM-pose ,(c)VitPose ,(d)YoloV8.

Our method performs excellently in terms of performance, achieving good accuracy with a relatively small number of training epochs. As shown in Fig. 5, the AP on the validation set steadily increases with more training epochs until it reaches a convergence point. Ultimately, our method achieved the highest AP value. Fig. 6 illustrates diverse predictions from different network models on the same image. The green markings represent the ground truth values, while the red markings depict the predicted results. It is evident that our method demonstrates high accuracy in predictions.

Additionally, the visualization in Figure 7 clearly shows that networks purely based on CNNs, such as ResNet and YoloV8, are often influenced by non-essential regions. These insignificant areas, highlighted with red boxes in the figure.

C. Ablation studies

Top-down multi-head heatmap We have validated the effectiveness of the top-down multi-head heatmap, a powerful

method with considerably higher costs compared to the MLP.

The number of channels in the feature maps produced by heatmap-based methods is closely tied to the number of key points, resulting in significantly larger feature maps compared to those directly regressed by the MLP for coordinate point output. However, this increase in cost leads to enhanced spatial generalization capabilities. In our experiments, both methods were examined, and the AAPE of the top-down heatmap-based approach increased by approximately 18% compared to the MLP-based method. Despite the increased cost, this effectively improves the accuracy of acupuncture point prediction.

V. DISCUSSION

Table II shows that our data covers all BMI categories. However, we have not thoroughly explored whether the model's accuracy is consistent across different body types. This limitation is largely due to the difficulty and constraints in data collection. Nonetheless, this is an intriguing task. We plan to increase the number of volunteers and conduct quantitative research on the impact of minority data types during inference. The inference accuracy, as indicated in Table III, is higher for males than females. This discrepancy is primarily attributed to the considerable influence of the dataset on the outcomes of deep learning. Comparatively, the dataset contains a significantly larger number of male data points, resulting in an imbalance in the male-to-female ratio within the training dataset. To enhance the localization accuracy for females, subsequent efforts should focus on augmenting the dataset by increasing the quantity of female data, thereby expanding the dataset's scale.

Fig. 4 and Table V present the results of two workflow approaches: direct prediction for all acupuncture points and inference prediction focusing only on benchmark acupuncture points, with the remaining acupuncture points inferred based on these benchmarks. Table IX summarizes the average errors for Partial acupuncture points calculated through benchmark-based inference. According to Table IV, evidently, the accuracy of this method is significantly contingent upon the precise localization of acupuncture points along the governor vessel meridian. Substantial positioning errors along the governor vessel meridian acupuncture points can lead to error accumulation at acupuncture points along the bladder meridian. For example, if there is a notable average error at the Shenzhu acupuncture point along the governor vessel meridian, it will consequently increase the average error in the prediction results for the Feishu and Fufen acupuncture points along the bladder meridian. This cumulative error effect has the potential to reduce the overall recognition accuracy. The results of Table IX are consistent with this inference.

In the keypoint recognition stage, the results directly obtained through MLP regression perform poorly compared to predictions based on heat maps. This is because the MLP network structure is overly simplistic, and incapable of handling more advanced image semantic features, resulting in poor direct regression outcomes. MLP simultaneously regresses all coordinate points without considering the potentiality of surrounding points as target points. Additionally, coordinates

TABLE VI
COMPARISON WITH EXISTING METHODS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

Method	AAPE	PCK-0.05	PCK-0.10
PVT [44]	10.54	0.81	0.96
Swin-pose [45]	11.81	0.75	0.98
RTM-pose [46]	11.96	0.88	0.99
VitPose [27]	10.01	0.89	1.00
YoloV8	14.877	0.87	0.96
Ours	9.29	0.93	1.00

TABLE VII
PERFORMANCE COMPARISON USING KEY METRICS: AP, AR, GFLOPs, PARAMETERS, AND FPS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD.

Methods	input size	FLOPs(G)	Params(M)	AP	AR	FPS
PVT [44]	192 × 256	4.919	28.188	0.734	0.770	19
Swin-pose [45]	192 × 256	6.373	32.785	0.660	0.740	25
RTM-pose [46]	192 × 256	2.099	16.109	0.82	0.85	13
VitPose [27]	192 × 256	18.596	90.011	0.820	0.85	14
YoloV8	192 × 256	187.6	48.41	0.60	0.60	62
Ours	192 × 256	151.799	73.89	0.856	0.879	13

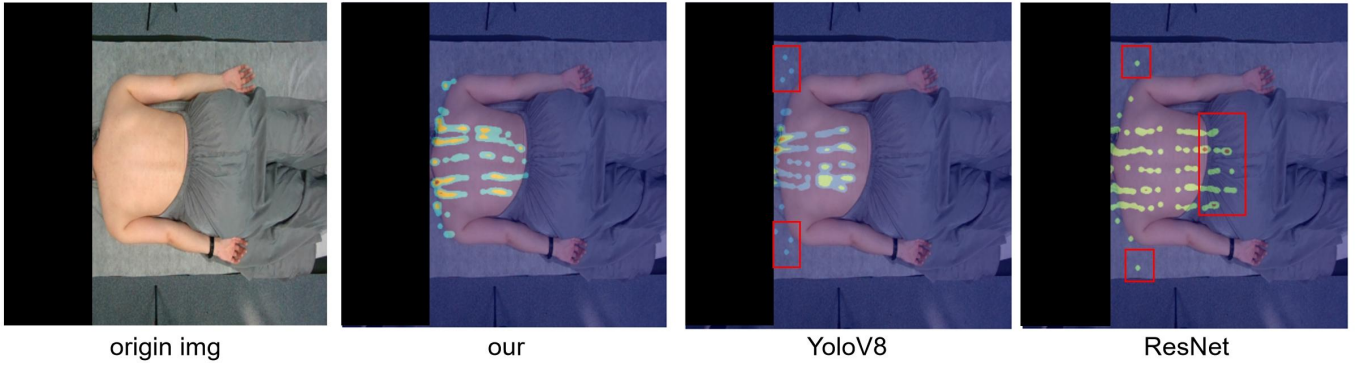


Fig. 7. Algorithm interpretability diagram

TABLE VIII
TOP-DOWN HEATMAP VS MLP

Workflow type	With Top-down heatmap	With MLP
Accuracy(AAPE)	9.29	12.50

regressed based on advanced image features suffer from a loss of precision during the upsampling process in quantization steps. Decoupling x and y and using one-dimensional heatmaps significantly reduces quantization errors. Furthermore, considering the local region around the target point helps avoid missing potential target points, effectively addressing the issue of low prediction accuracy.

In the actual detection and positioning process, the real-time video input often shows the phenomenon of fluctuation. This phenomenon is mainly due to the high sensitivity of the algorithm to small changes in the input image, resulting in rapid changes in the predicted acupuncture point locations. Even if the human eye sees very little change in the picture, the actual pixel level has been slightly changed, causing the prediction error between different frames.

As shown in fig.8, we employed the One Euro filter to refine

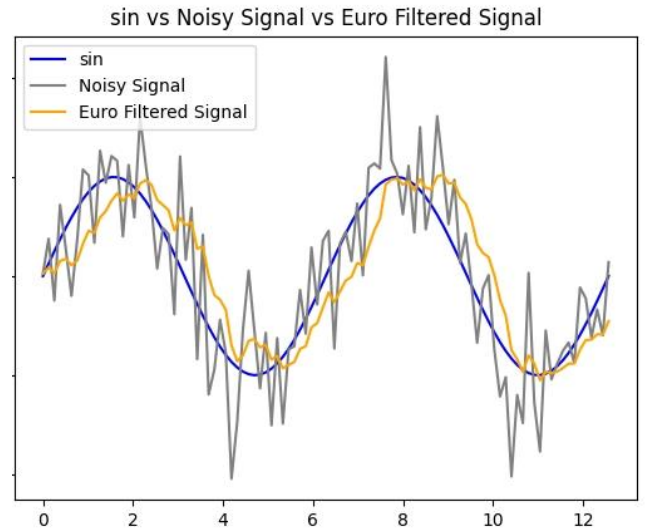


Fig. 8. Comparison of sin Signals with One Euro Filter.

TABLE IX
AVERAGE PREDICTION ERRORS FOR ALL ACUPUNCTURE POINTS ON THE BACK THROUGH BENCHMARK-BASED INFERENCE.

governor vessel meridian	AAPE	bladder meridian (inside)	AAPE	bladder meridian (outside)	AAPE
Dazhui	5.16	-	-	-	-
Taodao	8.38	Dazhu	8.19	-	-
-	-	Fengmen	29.059	Fufen	29.334
Shenzhu	39.637	Feishu	40.571	Pohu	41.326
-	-	Jueyin	38.036	Gaohuang	40.288
Shendao	34.182	Xinshu	34.647	Shengtang	36.357
Lingtai	31.829	Dushu	32.432	Yixi	33.530
Zhiyang	27.556	Geshu	28.220	Geguan	29.736
Jingsuo	27.700	Ganshu	28.686	Hunmen	29.512
Zhongshu	27.338	Danshu	28.482	Yanggang	29.295
Jizhong	21.538	Pishu	22.664	Yishe	24.952
-	-	Weishu	18.762	Weicang	20.392
Xuanshu	12.525	Sanjiao	15.222	Huangmen	17.491
Mingmen	10.751	Shenshu	13.053	Zhishi	15.335
Yaoyangguan	7.863	Qihai	12.536	Dachangshu	14.482

the predicted results through a first-order low-pass filter with an adaptive cutoff frequency. At low speeds, the low cutoff frequency enhances signal stability by minimizing fluctuations. Conversely, as the speed increases, the cutoff frequency rises to diminish lag. One Euro filter was used to filter the x and y coordinates of the prediction results of the Dazhui point in 40 consecutive frames, and the comparison before and after the filter was presented as shown in Fig.9. It is evident that the filtered prediction results exhibit reduced displacement, contributing to smoother performance in the video.

Limitation: 1) The training datasets were obtained in well-lit, clean laboratory environments. Consequently, AL-WFBS cannot effectively handle data with insufficient lighting, making it less robust against variations in lighting. 2) In real-time detection, the detection results theoretically should remain stationary without any movement. However, in practical applications, it has been observed that there is jitter in the detected acupuncture points when the subject is stationary. This phenomenon is speculated to be caused by the minimal changes in actual pixel values in computer vision, leading to variations in the predicted values. Despite the One Euro filter is applied to tackle this issue, the fluctuation still remains noticeable to human eyes.

Future work: In the future, we plan to gather depth information and deploy it on medical robots to enable robotic navigation. In addition, we are going to extend the back to other body parts, establish other body acupuncture point data sets, such as feet, arms, faces, etc., and eventually establish a whole-body acupuncture point database for the opening of the TCM metaverse.

VI. CONCLUSION

We propose an innovative keypoint prediction network based on transforms, with a unique approach using heatmaps for keypoint regression. During the keypoint prediction phase, our distinctive method enables efficient and precise keypoint localization. Specifically designed for the downstream task of predicting acupuncture points on the back, our approach excels in both speed and accuracy, it enables real-time recognition of acupuncture points on the back. This innovative approach injects new vitality into the development of intelligent traditional

Chinese medicine, providing robust support for achieving more intelligent and efficient TCM-assisted diagnosis.

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INSTITUTIONAL REVIEW BOARD STATEMENT

The study was conducted in accordance with the Declaration of Helsinki, and approved by the Ethics Committee of Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences (protocol code SIAT-IRB-230715-H0666 and date of approval: July 25, 2023).” for studies involving humans. It was also approved by the Ethics Committee of Xin-Huangpu Joint Innovation Institute of Chinese Medicine (protocol code 2023EC001 and date of approval: August 21, 2023).” for studies involving humans.

REFERENCES

- [1] E. S. Yang, P.-W. Li, B. Nilius, and G. Li, “Ancient chinese medicine and mechanistic evidence of acupuncture physiology,” *Pflügers Archiv-European Journal of Physiology*, vol. 462, pp. 645–653, 2011.
- [2] J. Mao, B. Zhu, and H. Wang, “Acupuncture-based treatments on cervical spondylosis of vertebral artery type,” *TMR Non-Drug Ther*, vol. 4, no. 1, p. 7, 2021.
- [3] J. J. Mao and M. M. Khanna, “Integrating acupuncture with western medicine in cancer treatment,” *Acupuncture and Moxibustion as an Evidence-based Therapy for Cancer*, pp. 313–328, 2012.
- [4] A. S. Epstein, K. T. Liou, S. A. Romero, R. E. Baser, G. Wong, H. Xiao, Z. Mo, D. Walker, J. MacLeod, Q. Li *et al.*, “Acupuncture vs massage for pain in patients living with advanced cancer: the impact randomized clinical trial,” *JAMA Network Open*, vol. 6, no. 11, pp. e2342482–e2342482, 2023.
- [5] S. Shah, L. Godhardt, and C. Spofford, “Acupuncture and postoperative pain reduction,” *Current pain and headache reports*, vol. 26, no. 6, pp. 453–458, 2022.
- [6] J.-E. Oh and S.-N. Kim, “Anti-inflammatory effects of acupuncture at st36 point: a literature review in animal studies,” *Frontiers in Immunology*, vol. 12, p. 813748, 2022.
- [7] M. Wang, W. Liu, J. Ge, and S. Liu, “The immunomodulatory mechanisms for acupuncture practice,” *Frontiers in immunology*, vol. 14, p. 1147718, 2023.



Fig. 9. The x and y coordinates of Dazhui predicted results before and after 40 consecutive frames of video were filtered using the One Euro filter.

[8] S. Liu, Z. Wang, Y. Su, L. Qi, W. Yang, M. Fu, X. Jing, Y. Wang, and Q. Ma, "A neuroanatomical basis for electroacupuncture to drive the vagal-adrenal axis," *Nature*, vol. 598, no. 7882, pp. 641–645, 2021.

[9] J.-G. Lin, P. Kotha, and Y.-H. Chen, "Understandings of acupuncture application and mechanisms," *American journal of translational research*, vol. 14, no. 3, p. 1469, 2022.

[10] N.-N. Yang, L.-L. Lin, Y.-J. Li, H.-P. Li, Y. Cao, C.-X. Tan, X.-W. Hao, S.-M. Ma, L. Wang, and C.-Z. Liu, "Potential mechanisms and clinical effectiveness of acupuncture in depression," *Current neuropharmacology*, vol. 20, no. 4, p. 738, 2022.

[11] Z. Huang, H. Yu, C. Zhang, and Y. Huang, "Development and application of computer vision," *Information and Computer*, vol. 32, no. 17, pp. 127–129, 2020.

[12] K. Huang, X. Zhao, Q. Li, and S. Hu, "Visual turing: The next step in the development of computer vision from the perspective of human-machine competition," *Journal of Graphics*, vol. 42, no. 3, p. 339, 2021.

[13] Y. Freund and R. E. Schapire, "A decision-theoretic generalization of on-line learning and an application to boosting," *Journal of computer and system sciences*, vol. 55, no. 1, pp. 119–139, 1997.

[14] C. Cortes and V. Vapnik, "Support-vector networks," *Machine learning*, vol. 20, pp. 273–297, 1995.

[15] R. Girshick, "Fast r-cnn," in *Proceedings of the IEEE international conference on computer vision*, 2015, pp. 1440–1448.

[16] W. Liu, D. Anguelov, D. Erhan, C. Szegedy, S. Reed, C.-Y. Fu, and A. C. Berg, "Ssd: Single shot multibox detector," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part I 14*. Springer, 2016, pp. 21–37.

[17] X. Fang and R. Yu, "Three-dimensional object recognition based on grid statistical model," *Computer and Modernization*, no. 4, pp. 24–28, 2014.

[18] C. Li, H. Wang, and B. Li, "Application of 3d model in sar image

automatic target recognition," *Aviation Engineering Progress*, vol. 8, no. 2, pp. 125–129, 2017.

[19] A. Toshev and C. Szegedy, "DeepPose: Human pose estimation via deep neural networks," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2014, pp. 1653–1660.

[20] A. Jain, J. Tompson, M. Andriluka, G. W. Taylor, and C. Bregler, "Learning human pose estimation features with convolutional networks," *arXiv preprint arXiv:1312.7302*, 2013.

[21] G. Papandreou, T. Zhu, N. Kanazawa, A. Toshev, J. Tompson, C. Bregler, and K. Murphy, "Towards accurate multi-person pose estimation in the wild," in *2017 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2017, pp. 3711–3719.

[22] M. Andriluka, L. Pishchulin, P. Gehler, and B. Schiele, "2d human pose estimation: New benchmark and state of the art analysis," in *Proceedings of the IEEE Conference on computer Vision and Pattern Recognition*, 2014, pp. 3686–3693.

[23] A. Bulat and G. Tzimiropoulos, "Human pose estimation via convolutional part heatmap regression," in *Computer Vision—ECCV 2016: 14th European Conference, Amsterdam, The Netherlands, October 11–14, 2016, Proceedings, Part VII 14*. Springer, 2016, pp. 717–732.

[24] Z. Cao, T. Simon, S.-E. Wei, and Y. Sheikh, "Realtime multi-person 2d pose estimation using part affinity fields," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2017, pp. 7291–7299.

[25] J. Li, S. Bian, A. Zeng, C. Wang, B. Pang, W. Liu, and C. Lu, "Human pose regression with residual log-likelihood estimation," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2021, pp. 11 025–11 034.

[26] K. Sun, B. Xiao, D. Liu, and J. Wang, "Deep high-resolution representation learning for human pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2019, pp. 5693–5703.

[27] Y. Xu, J. Zhang, Q. Zhang, and D. Tao, "Vitpose: Simple vision transformer baselines for human pose estimation," *Advances in Neural Information Processing Systems*, vol. 35, pp. 38 571–38 584, 2022.

[28] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, "Microsoft coco: Common objects in context," in *Computer Vision—ECCV 2014: 13th European Conference, Zurich, Switzerland, September 6–12, 2014, Proceedings, Part V 13*. Springer, 2014, pp. 740–755.

[29] Y. Li, S. Yang, P. Liu, S. Zhang, Y. Wang, Z. Wang, W. Yang, and S.-T. Xia, "Simcc: A simple coordinate classification perspective for human pose estimation," in *European Conference on Computer Vision*. Springer, 2022, pp. 89–106.

[30] F. Zhang, X. Zhu, H. Dai, M. Ye, and C. Zhu, "Distribution-aware coordinate representation for human pose estimation," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 7093–7102.

[31] G. Du, S. Lu, and H. Zhang, "Design of acupoint localization function for traditional chinese massage robot," *Science and Technology Bulletin*, vol. 27, no. 5, pp. 637–640, 2011.

[32] L. He, "Study on measurement and location of dangerous acupoints in head and neck," Master's thesis, TianJin University, 2012.

[33] H. Zhang, S. Lu, and G. Du, "Research on acupoint location and tracking based on template matching," *Science and Technology Bulletin*, vol. 27, no. 5, pp. 666–670, 2011.

[34] J. Gong, X. Shen, and Y. Liang, "Design of human acupoint locator," *Chinese Medicine Research*, vol. 31, no. 2, pp. 51–54, 2018.

[35] W. Cong, "The study on laser acupuncture robot visual acupoint searching method," Master's thesis, Beijing University of Posts and Telecommunications, 2020.

[36] C. Zheng, "Research on ar system of facial acupoint recognition based on deep learning," Master's thesis, Qingdao University of Science and Technology, 2022.

[37] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proceedings of the IEEE conference on computer vision and pattern recognition*, 2016, pp. 770–778.

[38] O. Russakovsky, J. Deng, H. Su, J. Krause, S. Satheesh, S. Ma, Z. Huang, A. Karpathy, A. Khosla, M. Bernstein *et al.*, "Imagenet large scale visual recognition challenge," *International journal of computer vision*, vol. 115, pp. 211–252, 2015.

[39] H. Zhang, F. Li, S. Liu, L. Zhang, H. Su, J. Zhu, L. M. Ni, and H.-Y. Shum, "Dino: Detr with improved denoising anchor boxes for end-to-end object detection," *arXiv preprint arXiv:2203.03605*, 2022.

[40] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint arXiv:1412.6980*, 2014.

- [41] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimeshein, L. Antiga *et al.*, “Pytorch: An imperative style, high-performance deep learning library,” *Advances in neural information processing systems*, vol. 32, 2019.
- [42] G. Casiez, N. Roussel, and D. Vogel, “1€ filter: a simple speed-based low-pass filter for noisy input in interactive systems,” in *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*, 2012, pp. 2527–2530.
- [43] X. Kuang, “Recognition and localization of acupoints on the human back based on deep learning,” Ph.D. dissertation, Nanchang University, 2022.
- [44] W. Wang, E. Xie, X. Li, D.-P. Fan, K. Song, D. Liang, T. Lu, P. Luo, and L. Shao, “Pyramid vision transformer: A versatile backbone for dense prediction without convolutions,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, October 2021, pp. 568–578.
- [45] Z. Xiong, C. Wang, Y. Li, Y. Luo, and Y. Cao, “Swin-pose: Swin transformer based human pose estimation,” in *2022 IEEE 5th International Conference on Multimedia Information Processing and Retrieval (MIPR)*. IEEE, 2022, pp. 228–233.
- [46] T. Jiang, P. Lu, L. Zhang, N. Ma, R. Han, C. Lyu, Y. Li, and K. Chen, “Rtmpose: Real-time multi-person pose estimation based on mmpose,” *arXiv preprint arXiv:2303.07399*, 2023.

Authors’ photographs and biographies are not available at the time of publication.